

Thick Seam Mining

The thickness of a coal seam is an important geological parameter during the selection of a suitable underground mining method for its efficient extraction. A coal seam of around 3 m to 4 m thickness is, generally, observed to provide normal working height for its efficient extraction and is suitable for most of the conventional mining methods. An increase in thickness of a coal seam makes it challenging for efficient underground mining and these seams are called thick.

Although there is no standard universal value of thickness to call a coal seam thick, different countries have varying limits (Singh, 2004) for categorizing thick coal seams. The adopted limit to call a coal seam thick in India is 4.8 m, which is a bit higher in comparison to the limits of other countries. Even after the adoption of this higher limit, over 60% of the total coal reserve (Information Bulletin, 2003) in the country, workable by underground mining methods, belongs to thick seams.

Ground control and Rock mechanics aspects of thick seam mining

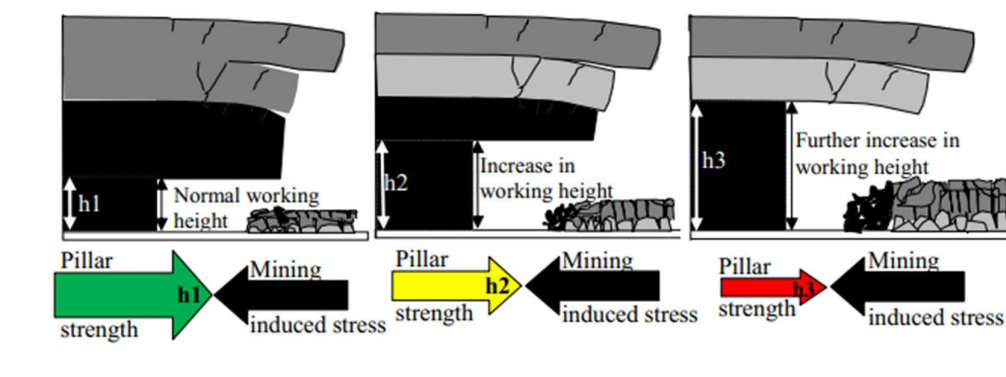


Fig. 1 Effect of mining height on pillar strength

- **W/H Ratio:** Referring to the above figure, it can be seen that the strength of the pillar decreases with the increase in pillar height for a given geo-mining condition. Additionally, it indicates that for a given width of the pillar, as we increase the height of the pillar, which alternatively lowers the W/h ratio of the working, it causes a reduction in pillar strength by lowering its structural stiffness against the roof loading for a given mining condition.
- **Natural Pressure Arch (NPA):**
 - The stress arch is regarded as the bearing structure during overlying strata movement. During the strata movement, the stress arch transfers a load of overlying strata to the surrounding rock around

the working face and forms the front and rear arch abutment with a stress-concentration zone in the coal seam and goaf, leaving a stress relief zone above the mined coal.

Load Distribution

from overlying strata

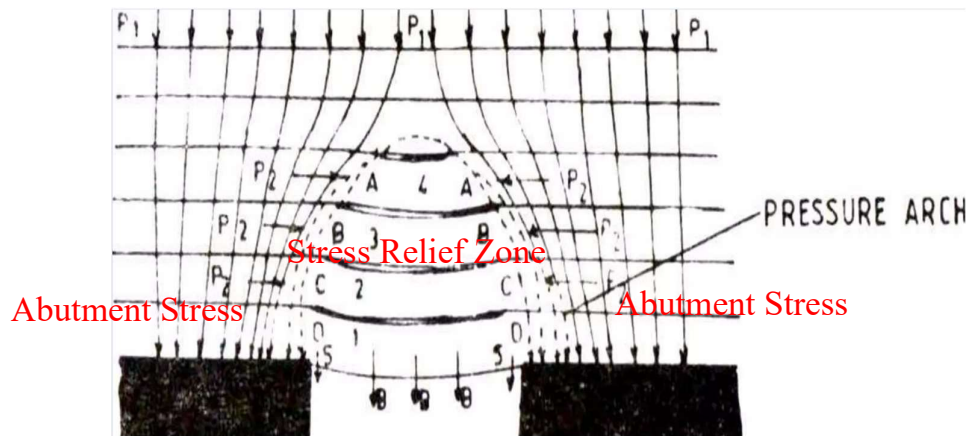


Fig. 2 Pressure arch formation mechanism

- From the field practices, it has been seen that with the increase in mining height/thickness, the NPA increases significantly causing higher stress concentration in the form of abutment stress surrounding the working.

- **Pillar Spalling**

- The higher the thickness of the coal seam, the lower is stiffness of the structure, which lowers the resistance of the coal seam to the deformation under a given load.

- ***en-masse* Caving**

- The higher the thickness of the mining, the greater the height of NPA in the goaf (Refer Fig. 2). In this process the volume of rock that may participate in caving will be more, causing high volume/*en masse* caving by throwing strong weight to the working area.

Longwall Mining

- **Effect of Face length on strata Fracturing and Weighting**
 - Initiation and propagation of failure in the main roof have a significant influence on loading at the face. The effect of longwall layout on failure and breaking process of the main roof through Kirchhoff's elastic plate theory.
 - The study suggested that the first crack always occurs on the top of the plate above the middle of the longer axis and extends along the longer side of the goaf area followed by the shorter side.
 - After these cracks connect each other, the plate structure of the main roof collapses in the form of “*” as it reaches its limiting value.
 - It has been observed that the fracture and weighting process of the main roof is a function of (a/b) where ‘a’ and ‘b’ are the lateral dimensions of the plate (Fig. 3). Weighting observed in conditions $(a/b) < 1$ is relatively severe as a substantial portion of the face remains under influence of the main roof during its bending.

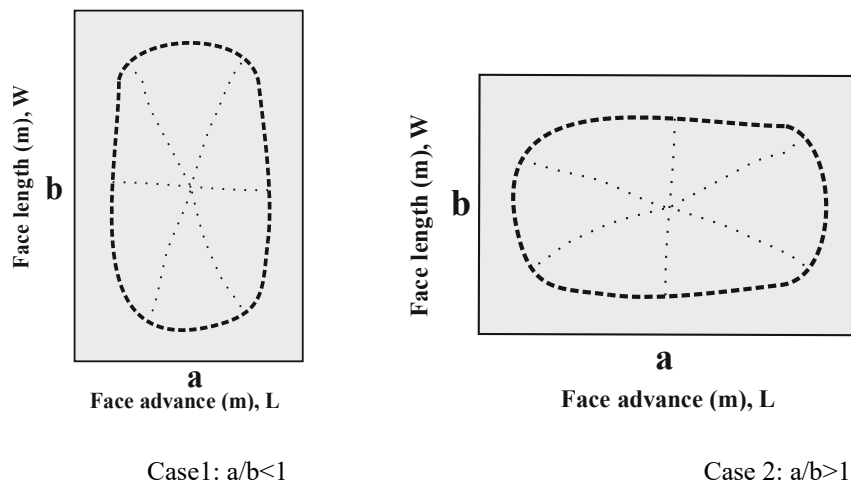


Fig. 3 Forms of main roof failure

- **Supporting Cycle and loading characteristics of Powered roof support:**
 - After a shield is set against the roof and floor, its load (or actually resistance to roof convergence and/or floor heave) does not remain constant at the setting load level. Rather it varies from period to

period as a result of interaction among roof strata, shield, and floor strata. There are distinct patterns of variations in shield resistance (or pressure) that can be used to evaluate the working conditions of the shield supports and the strata behavior.

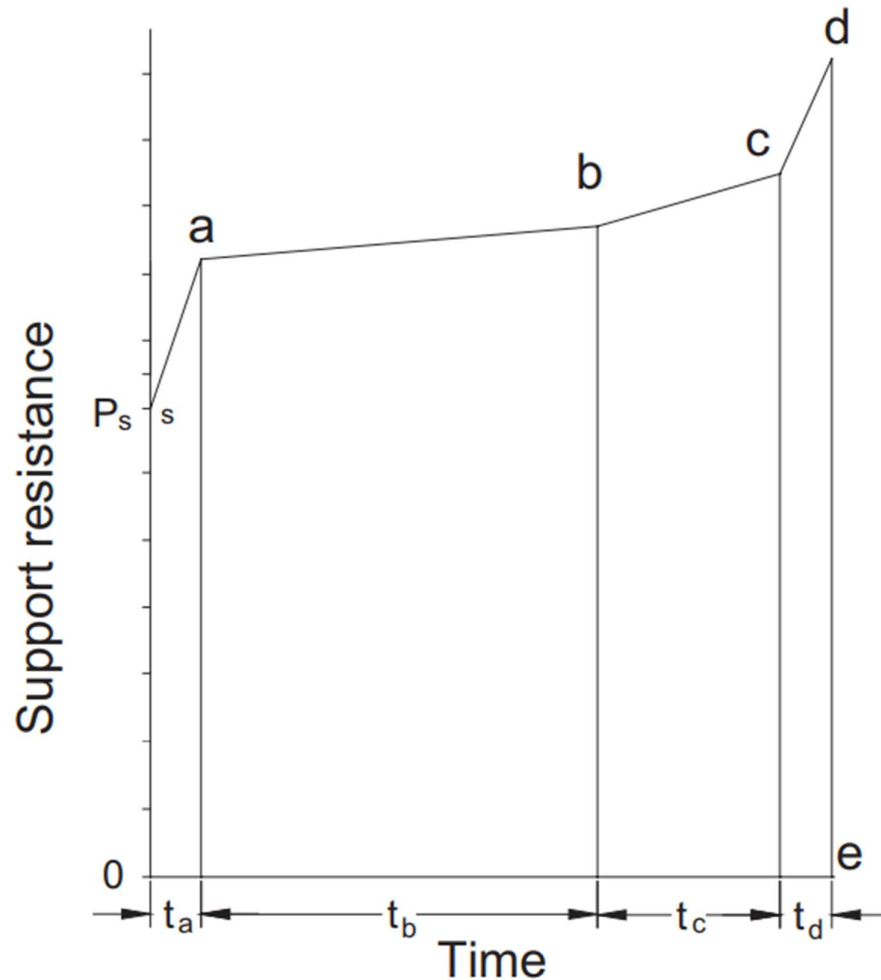


Fig. 4 Typical pressure change in a shield supporting cycle

- Figure 4 shows a typical form of variation of support resistance in a supporting (mining) cycle. Once the support has been advanced and reset, an initial setting load P_s , is rapidly achieved within a short time period, t_s , (10 sec–4 min). Thereafter, some may drop quickly various degrees depending on contact conditions (Fig. 4). This is the initial setting period (o–s). As the support starts to interact with the roof, the support resistance increases rapidly during time t_a until it reaches a relative equilibrium. This is the rapidly increasing resistance period (s–a).

- It is followed by a relatively stable loading period, a–b, during time t_b the shearer is not in the immediate vicinity of the support. In this relatively longer period of time, the roof loading is relatively stable. Notice this period is generally the longest in a mining cycle.
- When the drum cutting action of the shearer approaches and passes the support during time t_c , the roof loading increases as a result of the additional area of roof equivalent to the web width exposed. This is the cutting-influenced period, b–c.
- Finally, as the neighboring support is disengaged and advanced during time t_d , the roof load is suddenly transferred to the support, thereby rapidly increasing the support resistance in a short time period (generally less than 10 sec). This is another rapidly increasing resistance period, c–d.
- As the support is released for advancing, its resistance drops almost instantaneously to zero (d–e). Thus, a supporting (mining) cycle causes the support resistance or load (or pressure) variation from o through s, a, b, c, d, to e. Note P_s at s is setting pressure and d is final pressure. The difference between setting and final pressures is load increment.

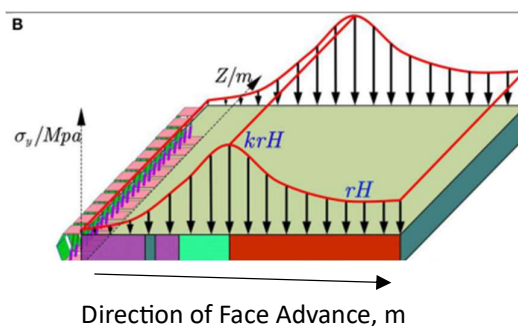
Ground Control Considerations

- **Abutment Stress Distribution about the longwall panel**

- The distribution of abutment stresses at the longwall face depends on the structural characteristics of the coal seam and the adjacent strata, mining depth, panel dimension, type and capacity of support etc. The magnitude of the abutment stress is usually related to overburden pressure and the function of mining depth, which is generally high because the majority of the longwall operates at higher depths.
- **Abutment Stress at the Face:**
 - **Across the Face (in the direction of face advance):**
 - Generally, after the extraction of the coal, the portion of mined coal and the amount of stress borne by this coal mass redistributes after the extraction and concentrates ahead of the working. The magnitude of these stresses is higher than the stress in situ and is called abutment stress. The abutment stress recorded

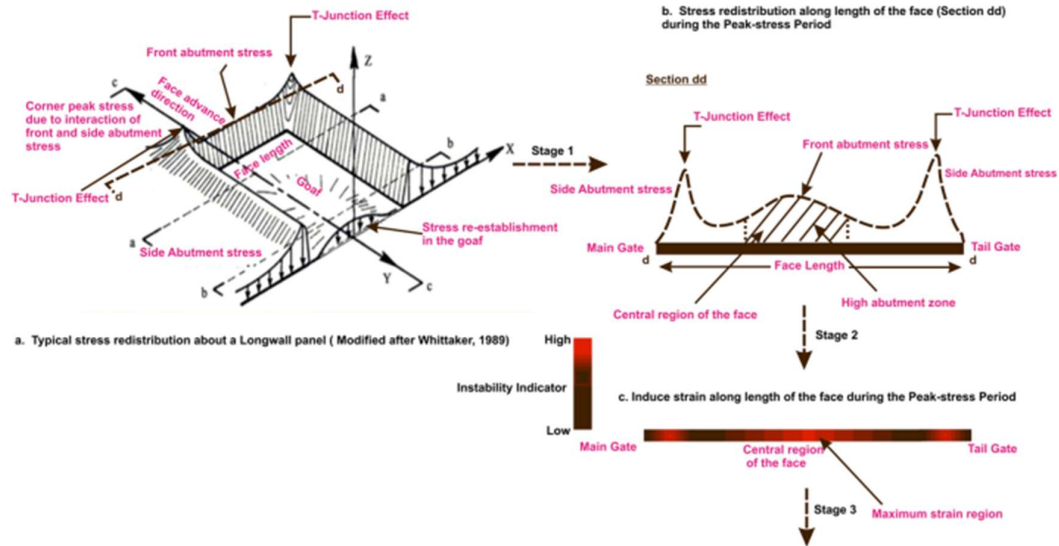
ahead of the longwall face is called front abutment stress.

- The red-lined curve in the below figures represents the typical distribution pattern of the front abutment stress ahead of the longwall face, in the direction of face advance. The peak abutment is generally noted at 2-5m ahead of the face in the Indian geo-mining conditions and depends on strata characteristics, depth of working, strength of the coal mass and the dimension of the face.



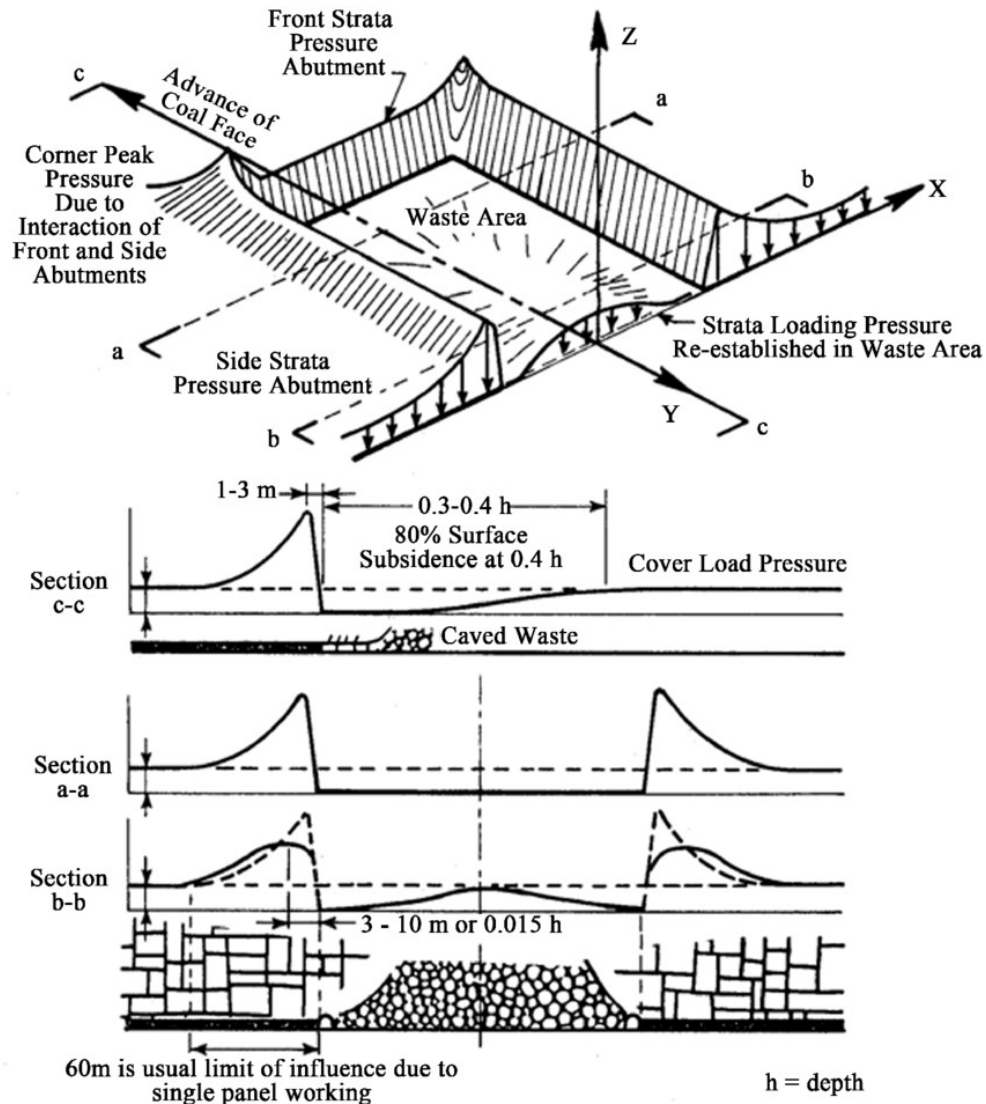
○ Along the Face Length

- Referring to the below figure (Fig. b), the typical front abutment stress distribution along the face length can be depicted. In a given condition, the T-Junction of the face i.e. gate end regions and central portion of the longwall face is often subjected to maximum stress conditions in progressive mining.
- With changing the dimension of the face/face length, the magnitude as well as the extent of maximum stress concentration region may vary.



- **At the Face corner:**

- The corner portion of the longwall face often experiences high abutment stress concentration due to the superimposition of the side abutment and front abutment stresses (Refer c-c section in the below figure).



○ In the goaf

- In the goaf, the central portion of the goaf experiences peak stress conditions due to the nature of compaction of gob material associated within this region.
- The goaf area just behind the face develops zero stresses owing to the zero compaction of the gob material or due to lack of contact of the gob material to the overlying roof (refer a-a section in the above figure).
- When the gob material touches the overlying strata, it starts to take the load from the roof and attain the compaction process. Under this process, it starts to develop some stress and reaches the maximum when 100% compaction is ensured (Refer b-b section in the above figure).

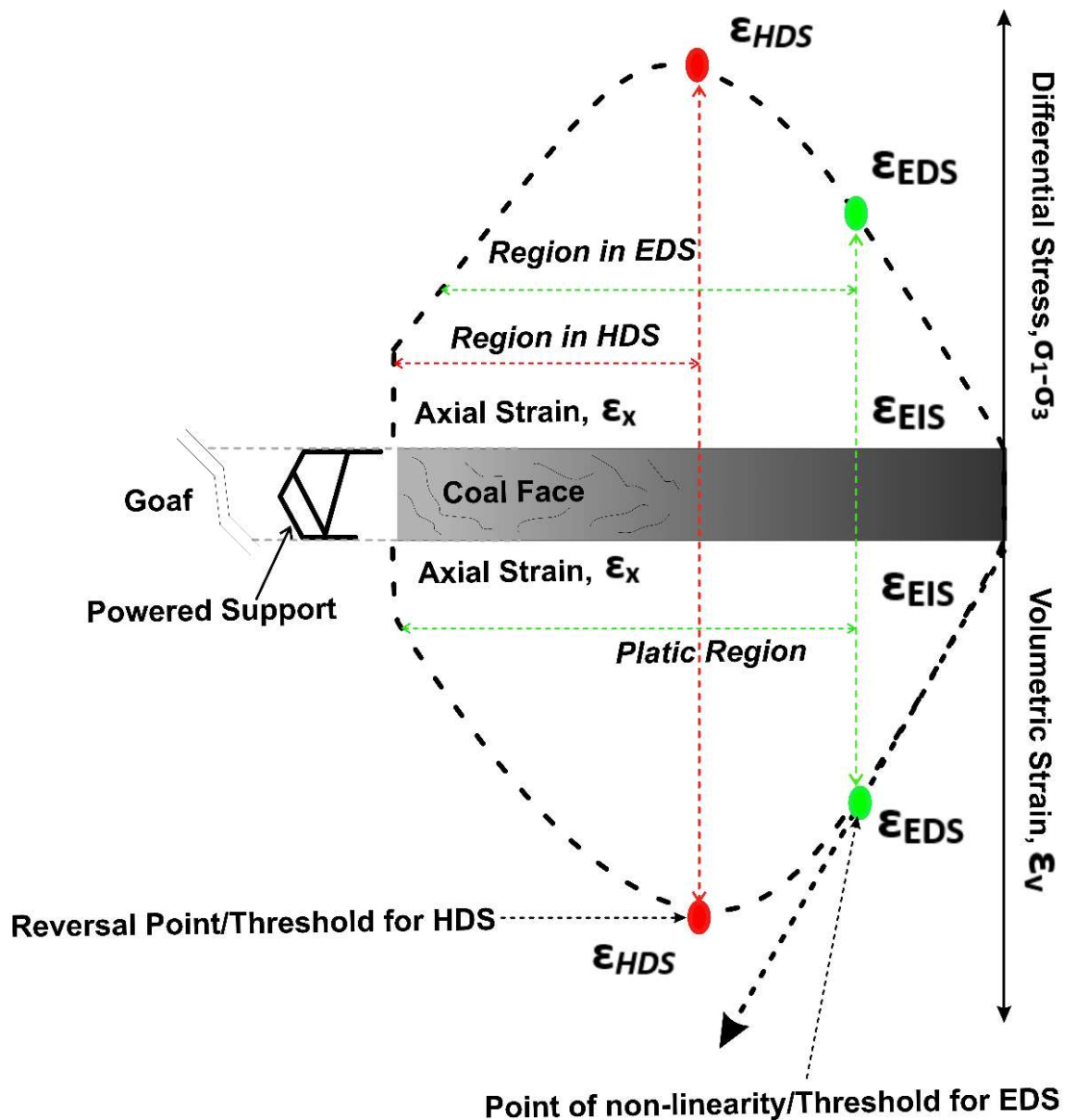
- The maximum stress that can be developed in the goaf is equal to the magnitude of in-situ stresses (the dotted line in the above figure represents the in-situ stress condition), the distance at which it attains this stress level is called cover pressure distance.

Typical Loading Conditions in the Longwall Operations

- A typical longwall system experiences three periodic loading conditions based on the failure characteristics of the main roof, termed as Normal, Peak-stress, and Weighting periods.
- The coal mass experiences a relatively stable stress condition in the normal period compared to that in the peak stress period owing to the propensity of strata bending, which contributes to the maximum stress concentration around the face.
- The magnitude of the front abutment is the maximum in the central region of a face, and this zone of peak abutment varies linearly with face length. For longer face length, the area of the maximum abutment zone at the center of the face could be very high, experiencing increased loading in comparison to the smaller face length.
- The state of loading can be catastrophic in the deeper longwall operations due to the higher stress environment.

Mechanism of Face Damage

- Characterization of the stress-induced damage at the longwall face is essential for quantifying the various forms of instabilities and their influence on structural integrity with progressive mining.
- Prior to the excavation, the coal mass typically behaves as an elastic media. The re-distribution of stresses caused by the excavation elevates the stress magnitude surrounding the rock mass and the coal face.
- Depending on the stress concentration intensity and associated strain rate, the longwall face attains various states of damage termed as *Highly Damage State (HDS)*, *Excavation Damage State (EDS)*, *Excavation Influence State (EIS)* and *Intact Region (IR)*.
- The states representing *HDS* and *EDS* qualify under plastic /irreversible damage conditions, while the *EIS* refers to the elastic condition associated with reversible stress and strain parameters upon unloading.

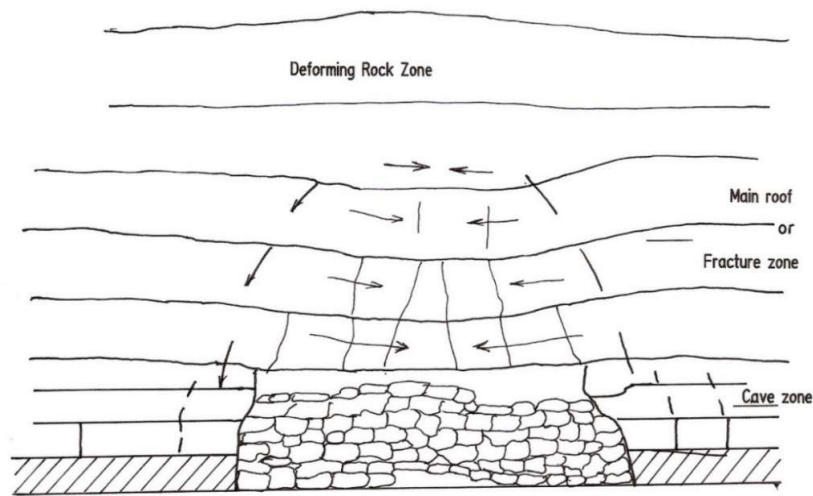


- The highly damaged zone (HDZ) attributes to the zones that may completely detach in the broken mass or slabs from the free face. This phenomenon is witnessed because of the interaction of maximum abutment and minimum confining stresses within such a region.
- The excavation damaged zone (EDZ) represents the fractured or failure mass formed around the opening, owing to the maximum governing stress within the structure. The irreversible change of rock mass properties is encountered in this zone.
- Evaluation of the depth of EDZ is crucial for a scientific understanding of the structural response of an opening in a given geological media and surrounding stress environment.

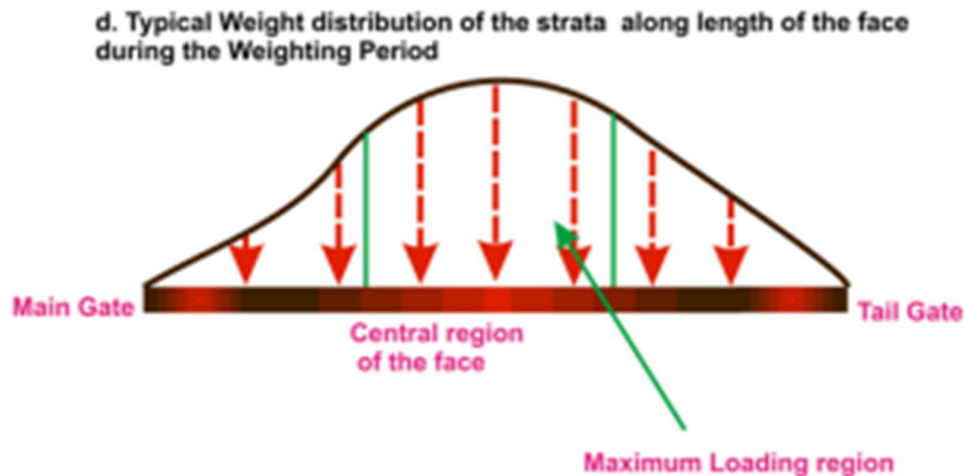
- The excavation influence zone (EIZ) refers to the zones formed immediately beyond the EDZ, where the changes in the rock mass properties are insignificant and reversible. The rock mass present in the EIZ is comparably stable and poses elastic behavior.
- Referring to the damage mechanics, under the space of axial strain vs. volumetric strain curve (refer to the above figure)), the deviation of the curve from the linearity is the indication of the formation of *Excavation Damaged State* and the reversal of the curve is the indication of the formation of *Highly Damaged State* at the coal face.

Roof weighting Mechanism

- Weighting at a longwall face occurs when a dominant immediate or the main roof above the goaf fails, triggering the lowering of the failed rock blocks and releasing energy due to the failure process. Until the failure occurs, the continuum strata above the goaf act as force-transmitting layers (Refer figure presented below), storing the strain energy released by the excavation. This accumulated energy is released at failure and subsequent caving of the strata. A part of the energy is directly transferred to the face supports and longwall face in this process.
- The occurrence of roof weighting at a large interval of face advance causes an increased loading of roof supports and a significantly increased peak convergence at the face. The stability condition of the roof also deteriorates in this process, as witnessed in terms of severe spalling of coal and formation of roof cavity along the face. Sometimes, such severe weighting also results in irreparable damage to hydraulic or mechanical components of the powered roof supports.



Different zones above a longwall face



- During the main fall, the amount of weighting caused at the coal face is called main weighting. The central part of the longwall face experienced maximum weighting compared to its gate end part (Refer the figure given above).
- This is primarily due to the propensity of strata bending along the length of the face, which is often experienced at higher side at the central part of the face length during main fall.

Face Spalling Mechanism

- Longwall face spalling is one of the most critical indicators for identifying the severity of strata control for a given mining condition.

It carries the repercussion of both the strata and the face damage mechanics and is regarded as the crucial research component in longwall geomechanics. The displacing characteristics of the coal mass are attributed to the depth of working and associated cyclic stress-strain variation, the intensity of strata weighting, the strength of the coal mass and the dimension of the longwall face.

- The variability in the formation of the HDZ region and load distribution of the strata during the main fall is detrimental to defining the spalling potential of the working. Among the various parameters, cover depth, geotechnical properties of the coal face, and massiveness vis-à-vis elasticity of the main roof state the HDZ region around the longwall face, while the strata deformation characteristics, span of caving and face dimension of the operation influence the weighting potential. The likelihood of severe face spalling is manifested with the higher face dimensions operated under massive roof conditions close to the working.

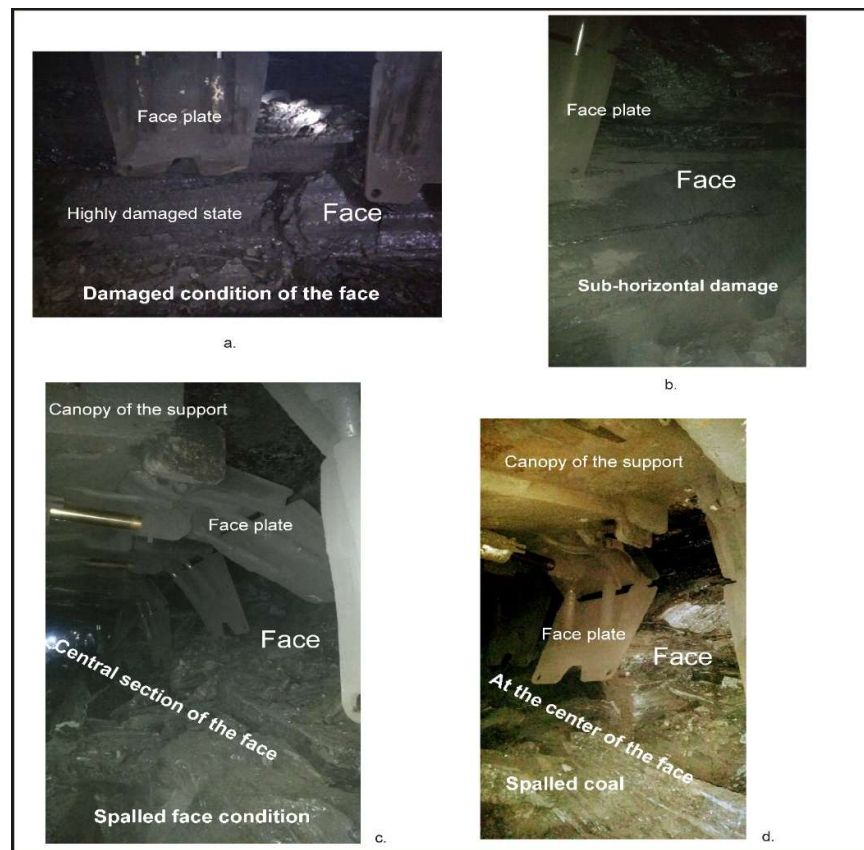


Fig. Face Damage and spalling mechanism recorded at the field